KEMAL ÖZYÖN



3rd-Year CTIS Student - Backend Systems & Applied AI

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3rd-year Computer Technology and Information Systems student (full scholarship, Bilkent University) focused on backend systems and applied AI. Hands-on experience building real-time communication platforms, desktop and mobile clients, and deployed web applications. Strong foundation in calculus and linear algebra; independently studying machine learning. Seeking a backend/AI-focused internship.

EDUCATION

B.Sc. in Computer Technology and Information Systems (Full Scholarship) *Sept 2023 - Jun 2028*

Bilkent University, Ankara

- Full merit-based scholarship; ranked 3,514th nationwide in the YKS university entrance exam (Quantitative Track).
- GPA: 3.24/4.00 (Last Semester: 3.55).
- Coursework: Data Structures & Algorithms, Backend & Frontend Development, Database Management Systems, Object-Oriented Programming, Linear Algebra, Calculus 1-2.
- Independent study: applied machine learning; preparing for AWS Certified Solutions Architect - Associate.

EXPERIENCE

Software Development Intern | KYANALİTİK (EczaneRapor) *Aug 7 - Sept 3, 2026*

- Developed FastAPI/WebSocket backends for collaborative editing and remote job execution, with JWT authentication, role-based access, async PostgreSQL, SQLAlchemy and Alembic.
- Built a CRDT-based collaborative editor with Yjs/pycrdt and Electron, React and TypeScript; developed React Native/Expo mobile clients for remote requests and screenshot delivery.
- Integrated Ollama to summarize web content with a local language model; implemented connection recovery, job scheduling, rate limits and URL/SSRF protections.
- Wrote unit, integration and end-to-end tests, containerized services with Docker, and built a Redis Pub/Sub prototype for WebSocket messaging across multiple server processes.

PROJECTS

Full write-ups & live demos: kemalozyon.com/projects

FastAPI Blog - *Python, FastAPI, PostgreSQL, SQLAlchemy, Alembic, JWT, AWS S3, Render*

- Built and deployed a full-stack blog serving server-rendered pages and a JSON REST API, with JWT auth and fully async PostgreSQL. Reviewed autogenerated Alembic migrations before production.
- Resolved production issues: moved email to an HTTP API when SMTP was blocked, migrated uploads to S3 for persistent storage, and fixed async lazy-loading errors with eager loading.

SustainMart - *Node.js, Express.js, MySQL, REST API, MVC*

- Built a two-sided marketplace backend (CTIS 256) with market/consumer roles, middleware-based authorization, two-step email verification, modular MVC structure and Multer image uploads.

Personal Portfolio Site - *Express, MongoDB, Mongoose, JWT, Cloudinary; React frontend*

- Built the backend for kemalozyon.com: authentication, content models, a JWT-protected admin dashboard and a Multer-to-Cloudinary upload pipeline. Directed and reviewed AI-generated React/Tailwind frontend work.

Demir Aslan - Digital Price Board - *Electron, Node.js, JavaScript, IPC*

- Built a desktop app used in a retail store to display real-time gold prices, with dual-window IPC synchronization and standalone Windows/macOS packages.

Bank Management System - *Java, OOP, Java Swing*

- Built a GUI banking application with multiple account types and modular separation of presentation and business logic.

SKILLS

Languages: Python, TypeScript, JavaScript, Java, C, C++, SQL
Backend: FastAPI, Express.js, Node.js, REST APIs, WebSocket, JWT, MVC
Databases & Messaging: PostgreSQL, MySQL, MongoDB, Redis Pub/Sub, SQLAlchemy, Mongoose, Alembic
Desktop & Mobile: Electron, React, React Native, Expo, Yjs/pycrdt
Cloud & DevOps: AWS S3, Render, Docker, Git/GitHub, GitHub Actions, Linux
AI & Testing: Ollama, LangGraph, OpenAI API, pytest, Vitest, Playwright